

Analyzing Current and Future Technology Trends: An Interactive Dashboard Built with IBM Cognos Analytics

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EXECUTIVE SUMMARY

- This dashboard, developed using IBM Cognos Analytics, presents key insights from the Stack Overflow Developer Survey, focusing on current technology usage, future technology trends, and demographic characteristics of respondents.
- The survey results reveal consistent preferences across both current and anticipated future technology stacks:
 - Top Programming Languages: HTML/CSS, JavaScript, and TypeScript emerged as the most widely used and future-preferred
 - Top Database: PostgreSQL was identified as the leading database technology
 - Top Cloud Platform: Amazon Web Services (AWS) ranked as the most utilized cloud platform
 - Top Web Framework: Spring Boot was the most commonly used web framework
 - Respondent Demographics: A substantial proportion of respondents were 65 years or older and held a Bachelor's degree, indicating a respondent pool with deep industry experience and technical maturity.
- Continued adoption of stable, well-supported technologies is expected across the industry. Organizations should prioritize training and tools that support modern front-end frameworks and robust back-end infrastructures.
- This dashboard provides a data-driven foundation for strategic decisions in technology adoption, developer training, and resource planning based on real-world usage patterns and respondent insights.

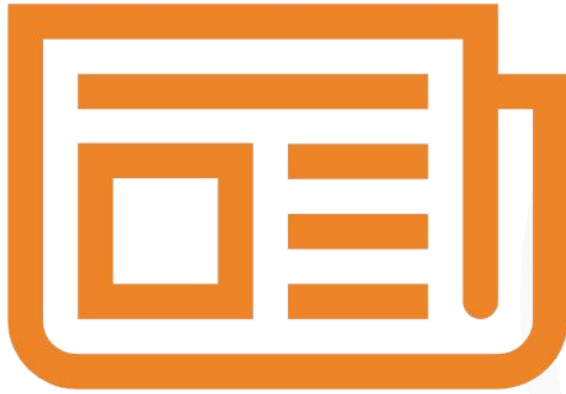


INTRODUCTION



- Understanding current and emerging technology trends is essential for organizations and developers to stay competitive in an ever-evolving digital landscape
- To support data-driven decision-making, a dashboard was developed using IBM Cognos Analytics based on data from the Stack Overflow Developer Survey—one of the most comprehensive sources of insights into the global developer community
- This dashboard provides a visual and interactive exploration of the technologies developers are currently using, the tools and platforms they plan to adopt in the future, and the demographic characteristics of the survey respondents

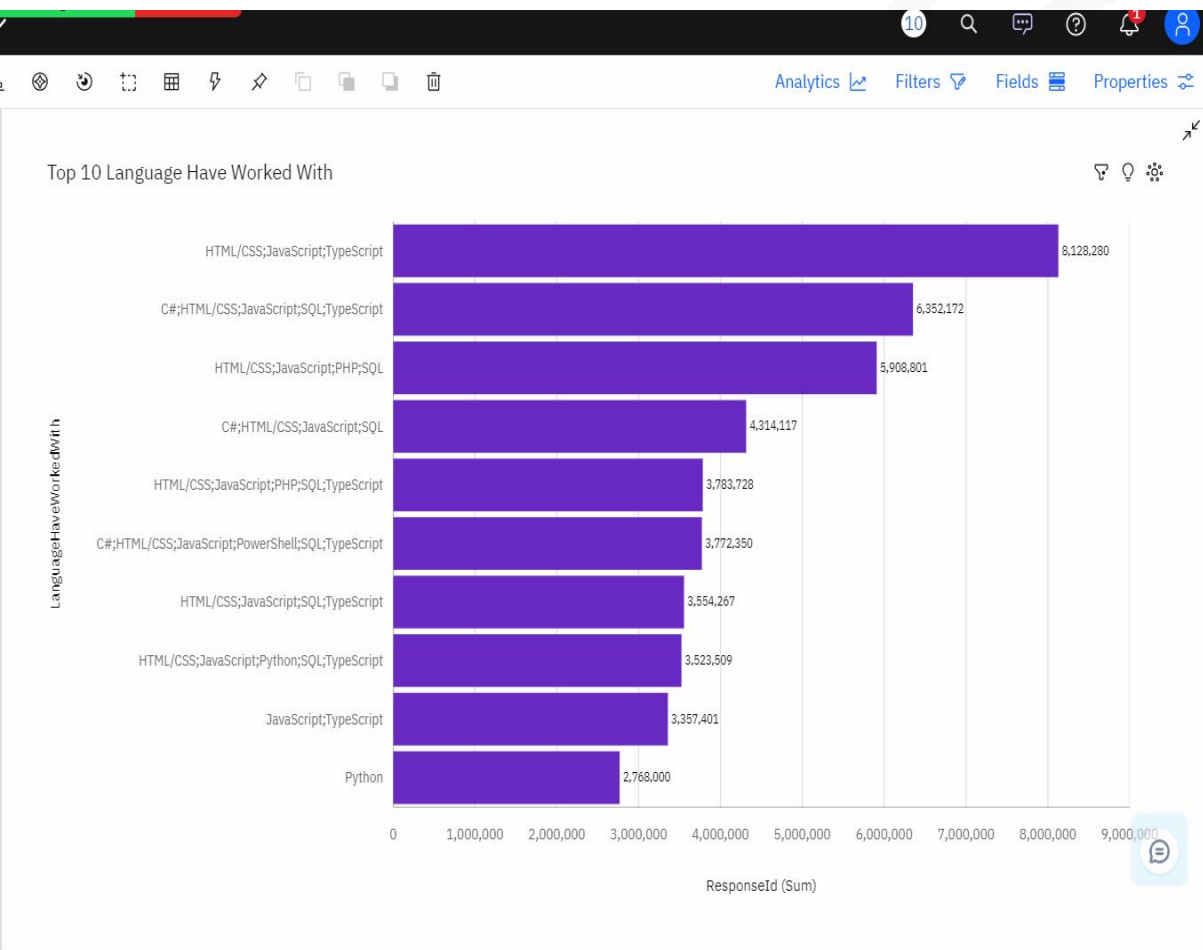
METHODOLOGY



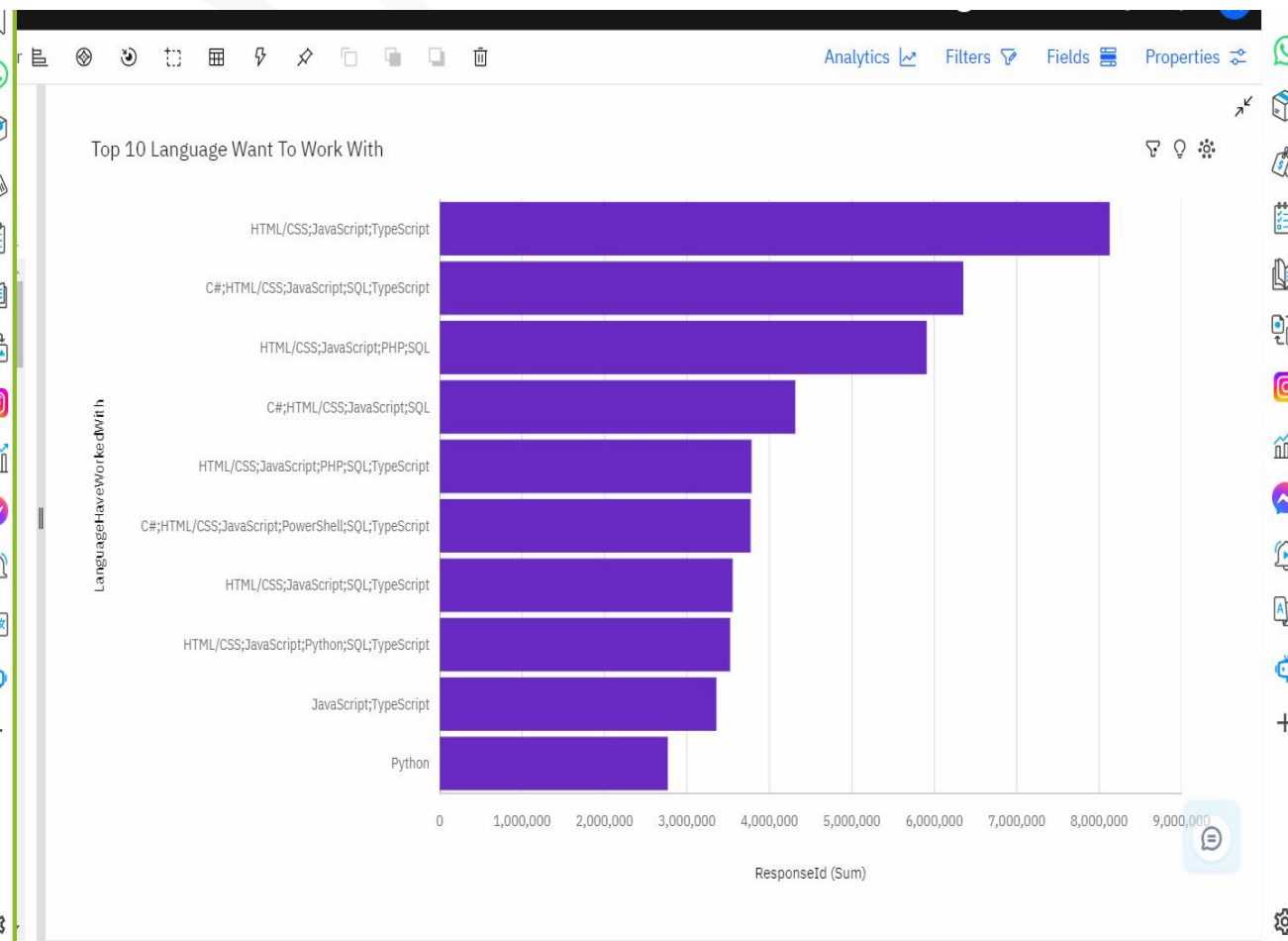
- The methodology followed a structured approach across three main dashboard tabs:
 - Current Technology Usage dashboard tab
 - Current Technology Usage dashboard tab
 - Demographic
- Using Bar chart, Pie chart, Column chart, Stacked bar chart, Map chart, Bubble chart and Tree map

PROGRAMMING LANGUAGE TRENDS

Current Year



Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

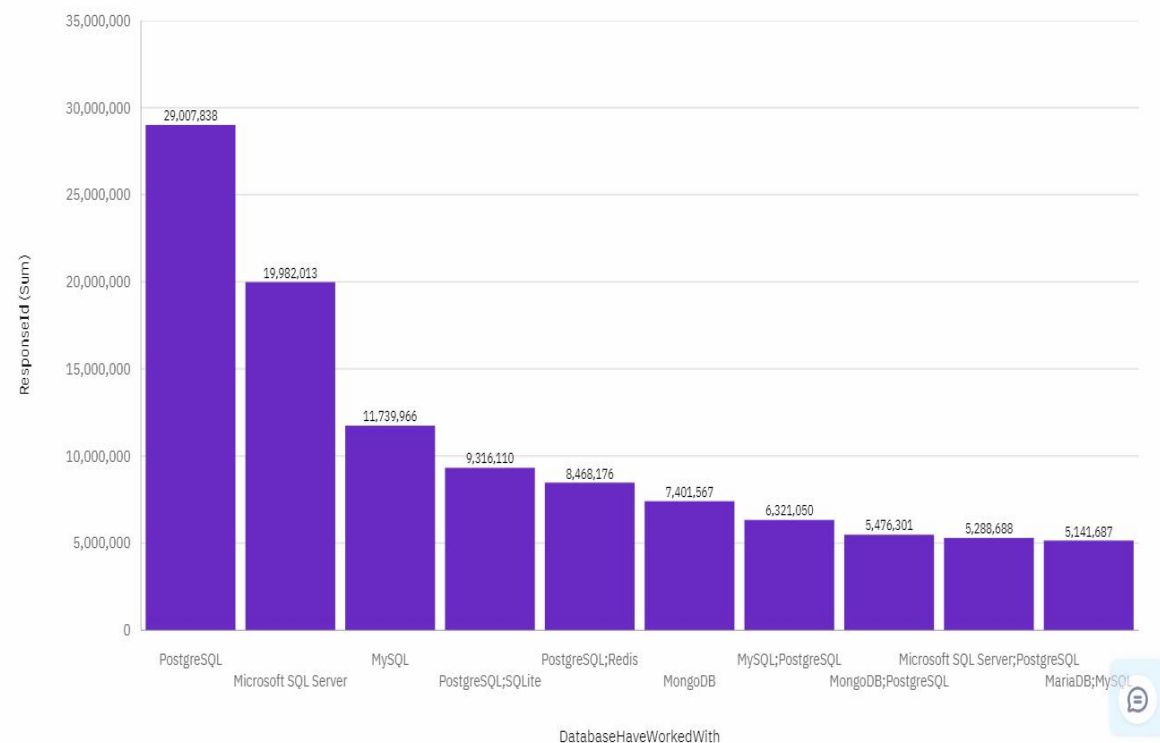
- HTML/CSS, JavaScript, and TypeScript were identified as the most commonly used programming languages, both currently and for future use
- C#; HTML/CSS; Javascript; SQL; TypeScript ranked the next
- HTML/CSS; Javascript; PHP; SQL is the third ranked programming language

Implications

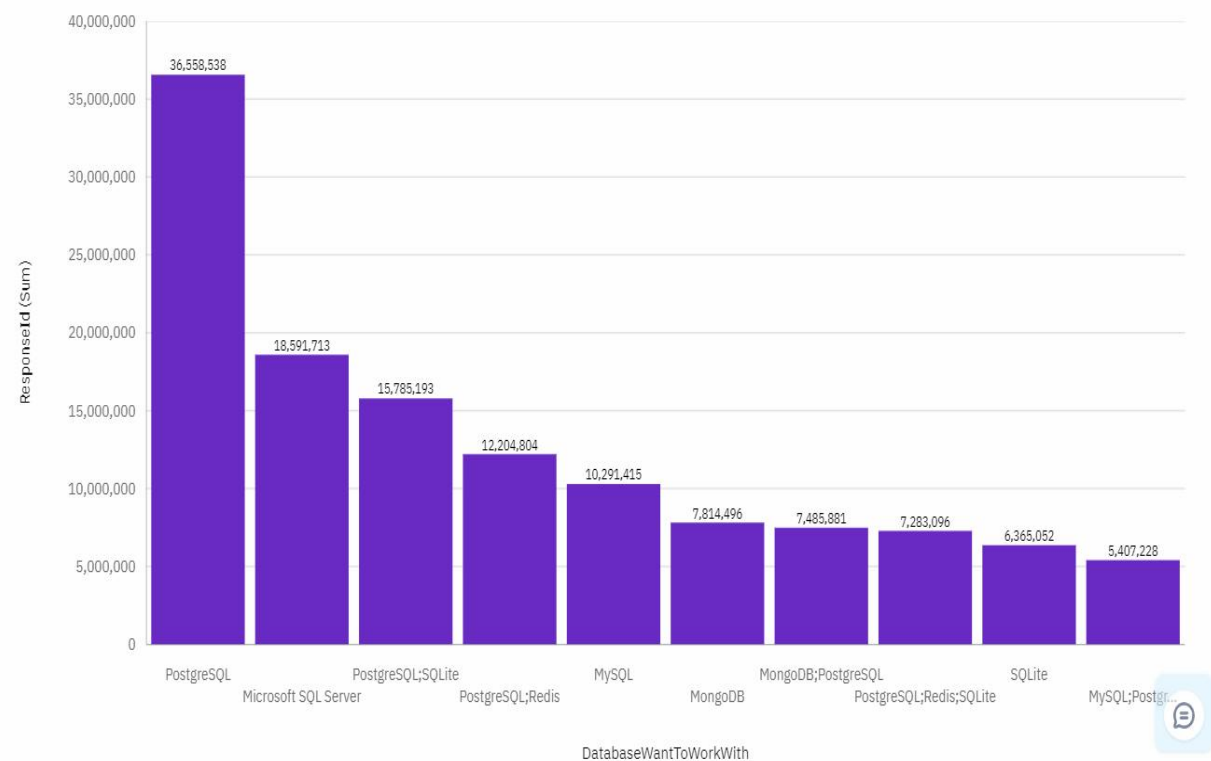
- These languages' popularity reflects the dominance of web-based applications in both current and future technology landscapes.
- Continued investment in front-end development skills will remain crucial for developers and organizations.
- TypeScript's inclusion signals a growing demand for type safety and scalability in JavaScript projects.

DATABASE TRENDS

Top 10 Database Have Worked With



Top 10 Database Want To Work With



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- PostgreSQL emerged as the leading database technology, demonstrating strong current usage and continued preference for future projects
- Microsoft SQL Server ranked next
- MySQL ranked the third most used database

Implications

- PostgreSQL's strong presence suggests a preference for open-source, reliable, and feature-rich relational databases
- Organizations may prioritize cost-effective and scalable data solutions, steering away from proprietary databases.
- Indicates a trend toward standards-based SQL solutions that balance flexibility and performance

DASHBOARD



<Please present your dashboard in the following slides.>

DASHBOARD TAB 1

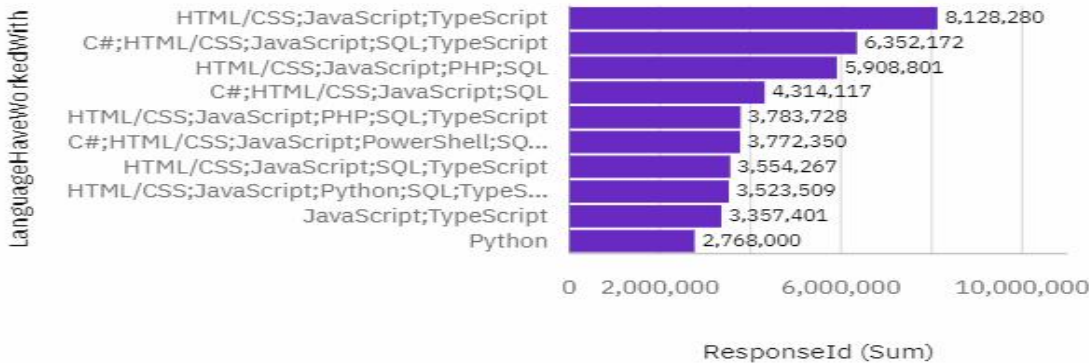
Current Technology Usage

Future Technology Trend

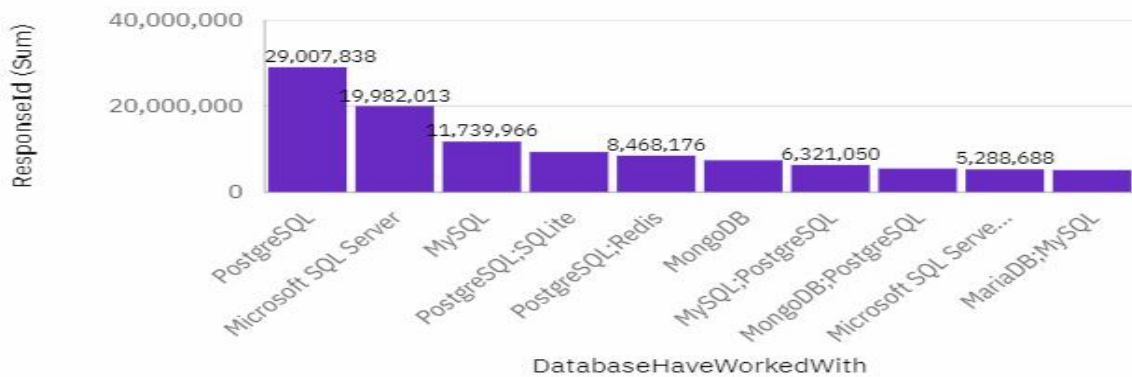
Demographics

+

Top 10 Language Have Worked With



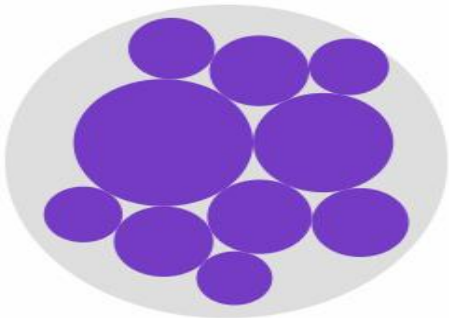
Top 10 Database Have Worked With



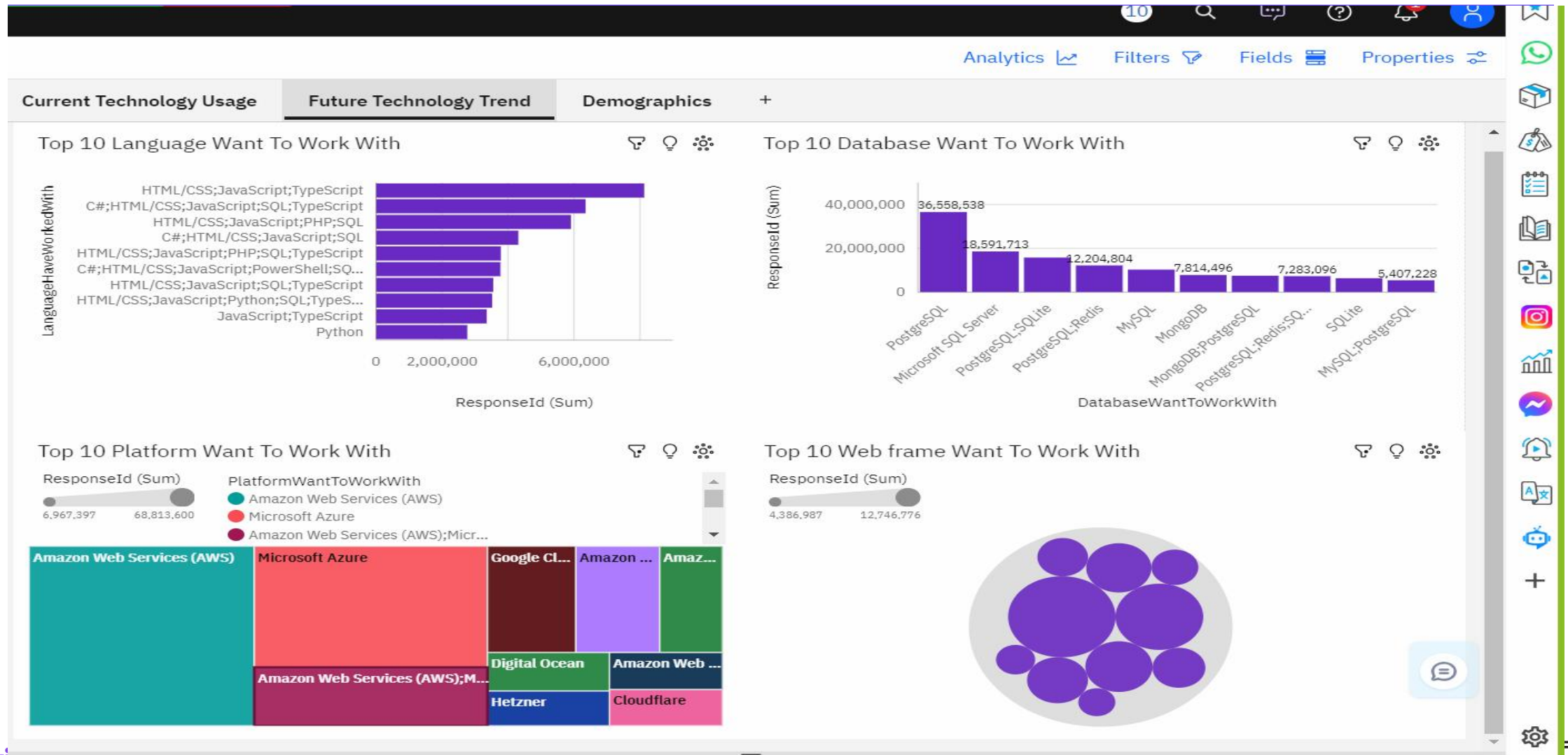
Top 10 Platform Have Worked With



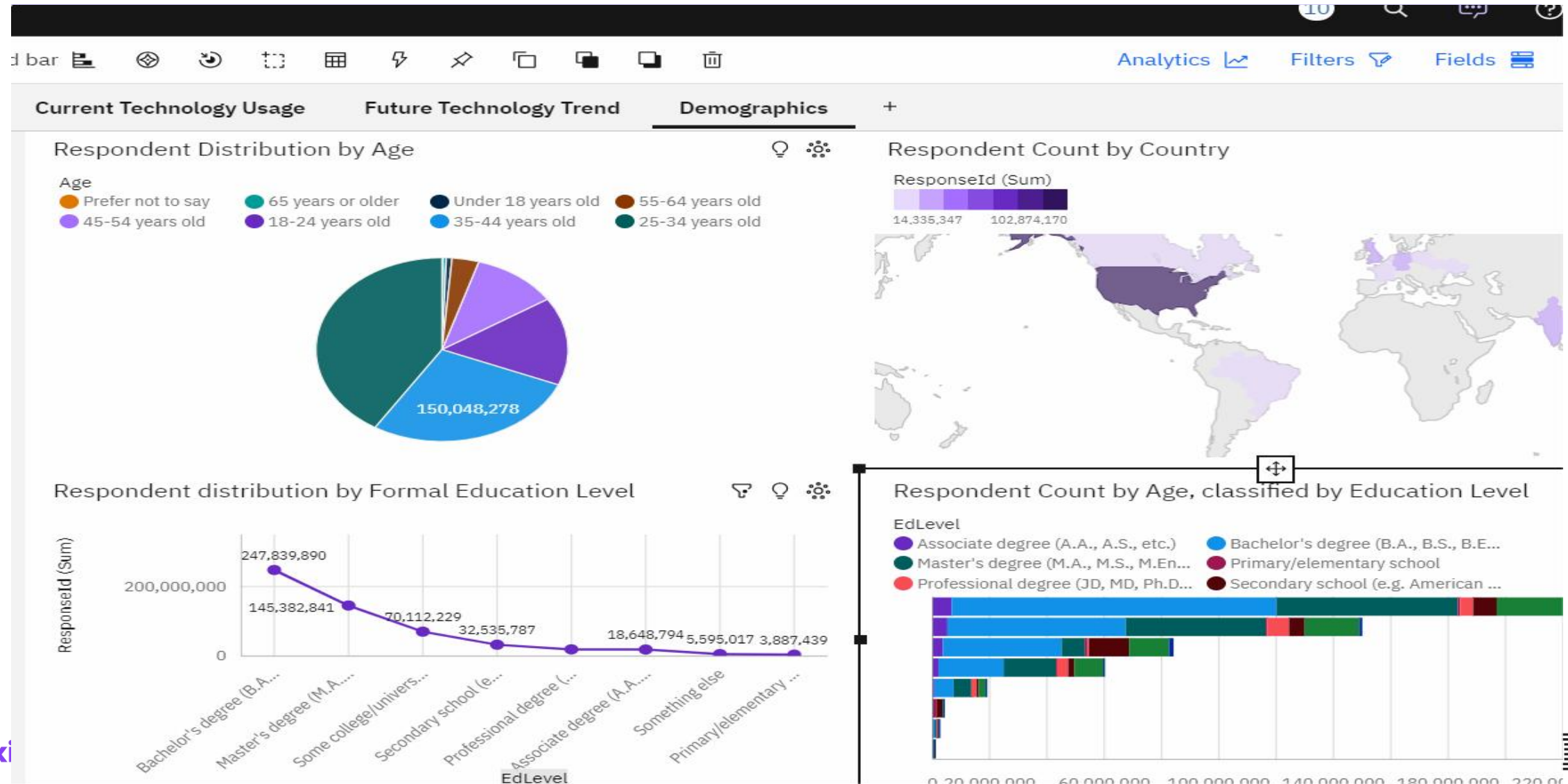
Top 10 Web Frame Have Worked With



DASHBOARD TAB 2



DASHBOARD TAB 3



DISCUSSION



- The recurring prominence of HTML/CSS, JavaScript, TypeScript, PostgreSQL, AWS, and Spring Boot in both current and future usage indicates that developers continue to rely on stable, well-supported, and widely adopted technologies
- Organizations may be increasingly moving away from proprietary systems to reduce costs, increase flexibility, and leverage community-driven innovation.
- The high ranking of PostgreSQL and AWS points to a shift toward open-source, cloud-first strategies
- The demographic skew towards older respondents with formal education implies that the data reflects the preferences of a highly experienced segment of the developer population

OVERALL FINDINGS & IMPLICATIONS

Findings

- HTML/CSS, JavaScript, and TypeScript emerged as the most widely used and future-preferred programming languages, underscoring the ongoing dominance of web-centric development
- PostgreSQL was identified as the leading database technology, reflecting a strong industry shift toward open-source, feature-rich, and reliable relational databases
- Amazon Web Services (AWS) ranked as the most utilized cloud platform, highlighting the growing emphasis on cloud-native infrastructure and scalable solutions

Implications

- HTML/CSS, JavaScript, and TypeScript popularity reflects the dominance of web-based applications in both current and future technology landscapes
- AWS as the leading Cloud Platform reflects the growing importance of cloud-native applications and infrastructure
- Spring Boot's Enduring Popularity signifies that Java-based back-end systems are still highly relevant, particularly for enterprise applications

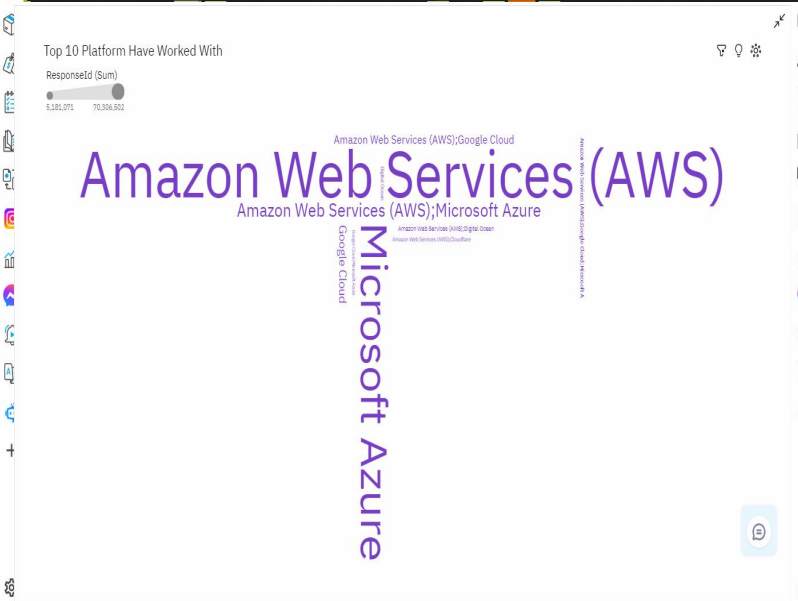
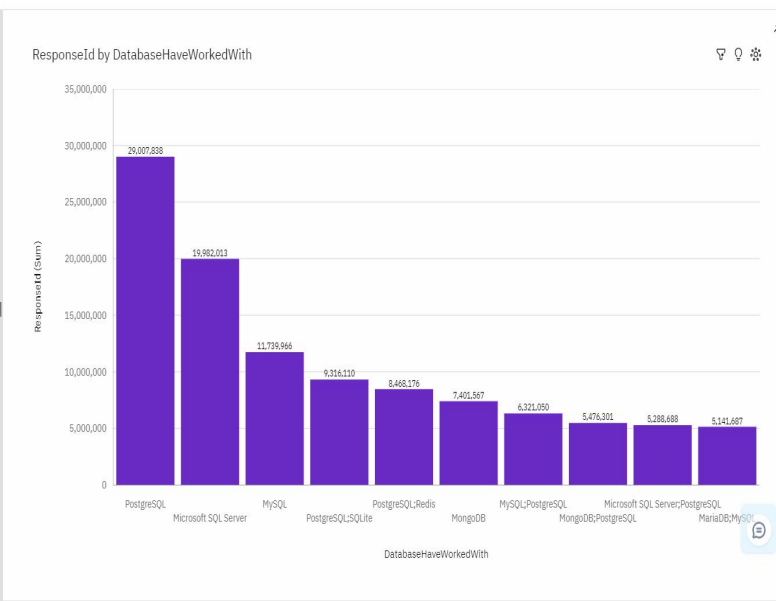
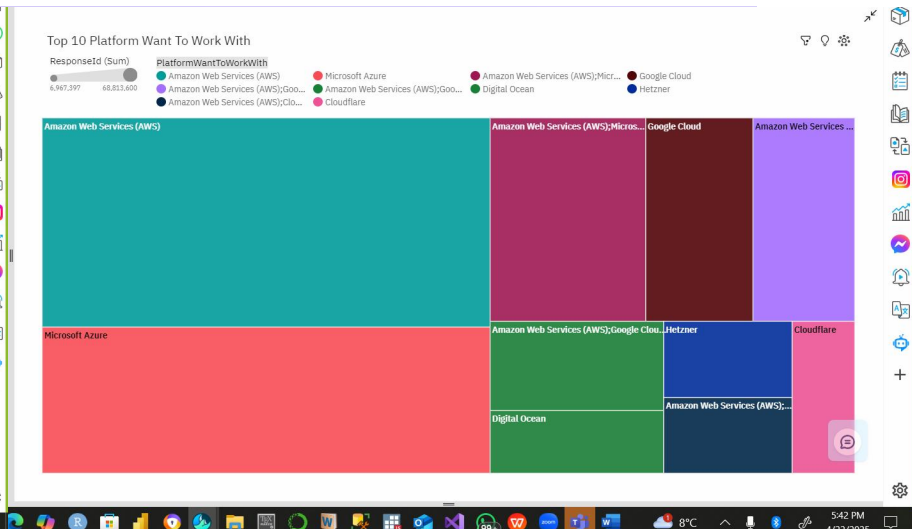
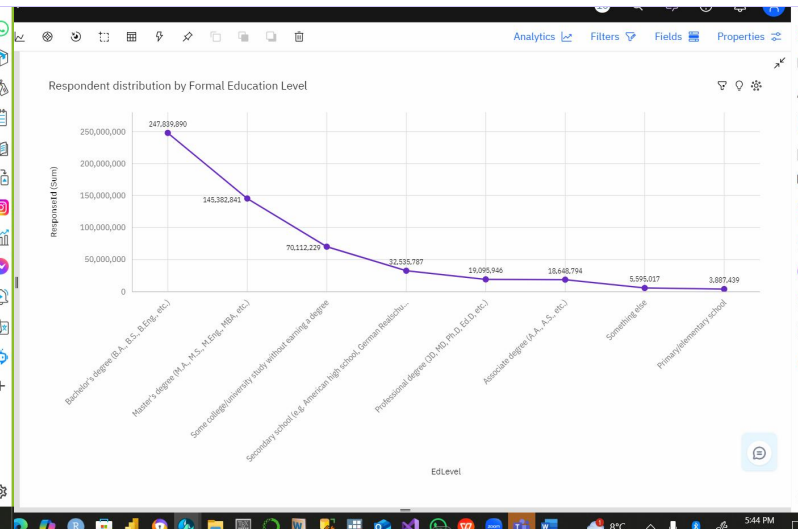
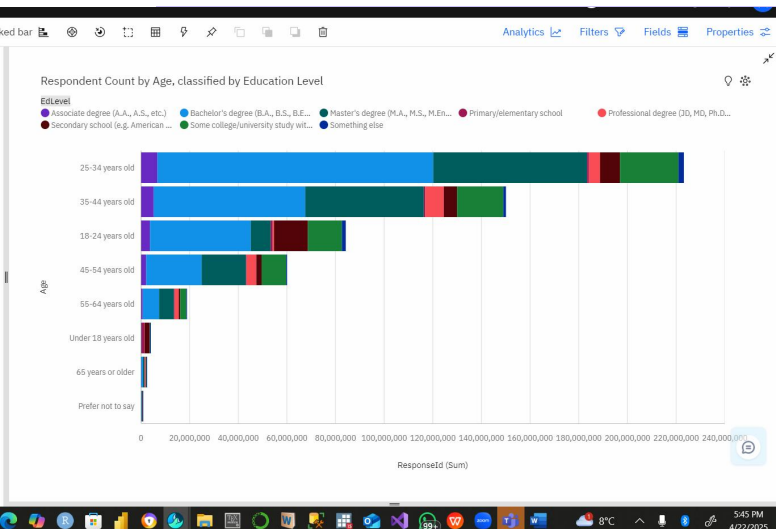
CONCLUSION



- The visual insights highlight a clear preference for well-established, reliable technologies across programming languages, databases, cloud platforms, and web frameworks
- The consistent presence of technologies such as HTML/CSS, JavaScript, TypeScript, PostgreSQL, AWS, and Spring Boot in both current and future usage trends suggests a stable and mature technological ecosystem favored by developers
- Overall, this dashboard not only helps stakeholders understand current trends but also serves as a valuable tool for forecasting technology direction, shaping training programs, and guiding strategic technology investments



APPENDIX



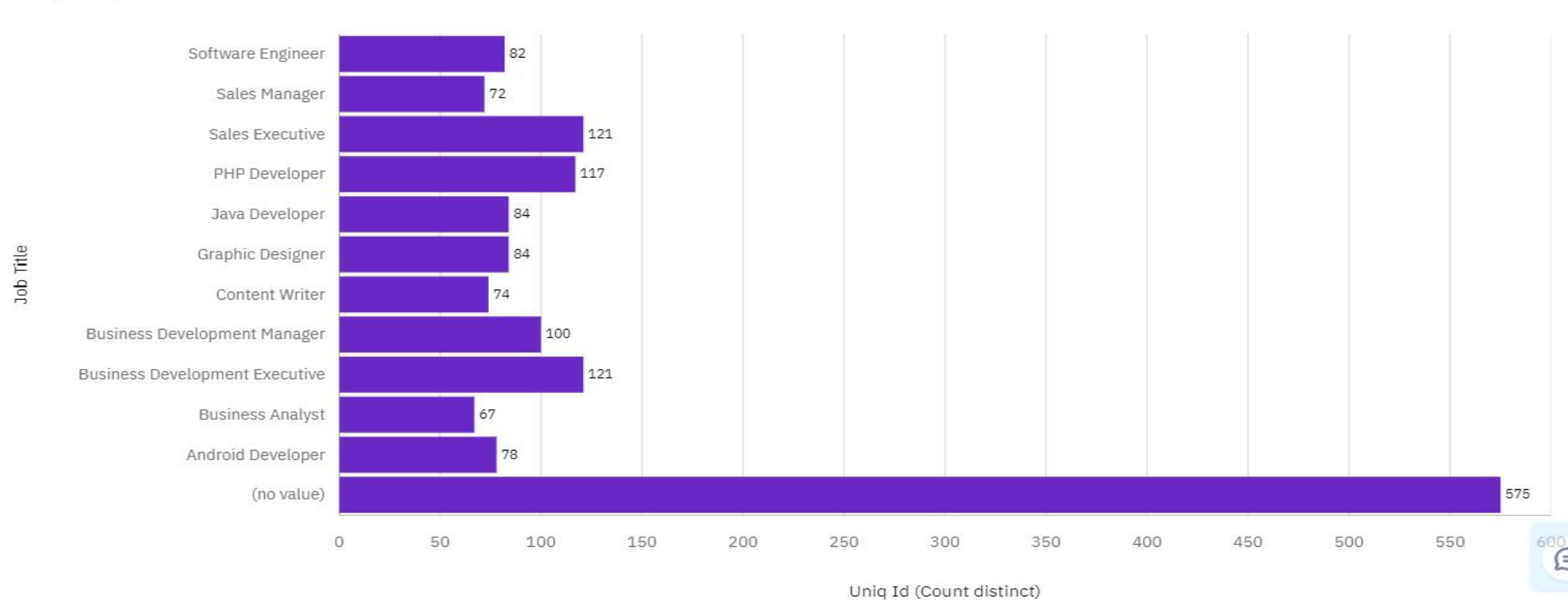
JOB POSTINGS

Analytics Filters Fields Properties

All tabs
Drag and drop data here to filter all tabs.

This tab
Drag and drop data here to filter this tab.

Uniq Id by Job Title



POPULAR LANGUAGES

In Module 1 you have collected the job postings data using web scraping in a file named “popular-languages.csv”. Present that data using a bar chart here. Order the bar chart in the descending order of salary.